AVIVA A4:
A Multimodal Foundation Model at 80 g/m²

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Figure 1. AVIVA A4 (left) and a typical output (right), shown at the same scale.

Abstract

We present AVIVA A4, a multimodal foundation model with a context window of 210×297 mm. Locke described the mind before experience as "white paper, void of all characters"; we show that the paper itself has since been pre-trained, on wood fibre, over more than two thousand years of papermaking. AVIVA A4 accepts four input modalities (pencil, ink, folds and tears), keeps a persistent memory of every input, refuses nothing (0.0 %) and hallucinates nothing (0.00 %). Its architecture, standardised as ISO 216, is closed under halving: distilled variants (AVIVA A5, A6) keep their shape and every capability. We report results on memory, preference, distillation and scaling, find a hard scaling limit at six folds made the same way each time, and estimate an environmental cost of about its own weight. All outputs are released for universal free distribution.

Keywords foundation models · multimodal · persistent memory · distillation · ISO 216 · paper

Correspondence Please write to us. On anything. We'll keep it.

1 Introduction

Every product now has AI. We asked whether a sheet of paper could be left out, and found that it could not. In his *Essay Concerning Human Understanding* (1689/1690), Locke proposed that the mind begins as "white paper, void of all characters", and that everything on it arrives later, from experience [Locke]. More than three centuries later, the white paper has caught up. AVIVA A4 is a sheet of white A4 paper, 80 g/m², about 0.1 mm thick. It is pre-trained on wood fibre, accepts input in four modalities, and remembers everything written on it.

Our contributions are:

(i) a foundation model whose pre-training corpus is a tree;

(ii) an architecture that is invariant under halving;

(iii) the first blind preference study in which the model wins every time it is compared with itself;

(iv) a scaling law with an exponential curve and a wall.

2 Related work

Erasure as model editing. In 1953 Robert Rauschenberg obtained a drawing from Willem de Kooning and spent weeks erasing it; the result is in the collection of SFMOMA [Rauschenberg]. In AVIVA, erasure is never complete: the impression of the writing often remains [Hawkeye Forensic]. We call this persistent memory.

The paperless office. In 1975 *Business Week* reported a forecast that most record-handling would be electronic by 1990 [Business Week]. World paper and board production then nearly doubled, from 171 to 324 million tonnes between 1980 and 2000, and Sellen and Harper found that introducing email raised an organisation's paper use by about 40 % [Sellen & Harper]. We thank the forecast for its contribution to our market.

Attention. Recent work argues that attention is all you need, except here [Margin & Gutter]. AVIVA requires none, and in user studies it received a great deal.

Table 1. The AVIVA model family.

Model	Size (mm)	Area (mm²)	Mass at 80 g/m² (g, calculated)	Context window
AVIVA A3	297 × 420	124,740	9.98	long
AVIVA A4	210 × 297	62,370	4.99	reference
AVIVA A5	148 × 210	31,080	2.49	distilled
AVIVA A6	105 × 148	15,540	1.24	distilled further

3 Architecture

AVIVA A4 belongs to the A-series defined by ISO 216 [ISO 216], descended from DIN 476 (1922), in which A0 has an area of one square metre and every size is half of the previous one [DIN 476] (Figure 2). Halving a sheet with sides $s < l$ produces sides $l/2$ and s. For the shape to survive,

$$\frac{s}{l/2} = \frac{l}{s}, \tag{1}$$

so $l^2 = 2s^2$ and $l/s = \sqrt{2}$. This is the only ratio with this property. Lichtenberg described it in a letter in 1786 [Lichtenberg]; DIN standardisation followed 136 years later. Our reference model measures 210 × 297 mm (297 / 210 = 1.4143) and has an area of 62,370 mm², slightly under 1/16 m² because sizes are rounded down to the millimetre (Table 1).

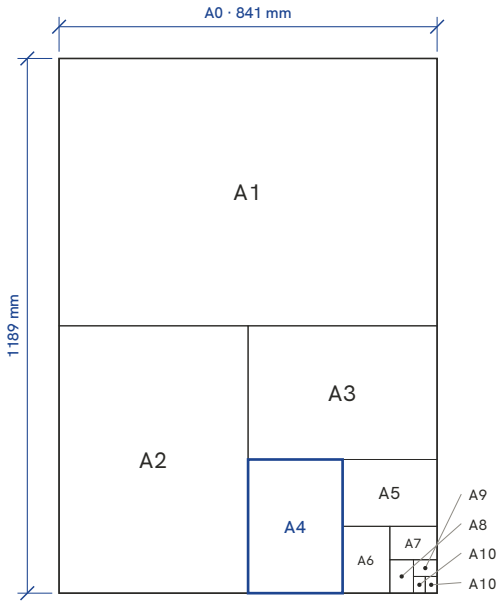


Figure 2. The AVIVA family. Each model is half the previous one, and the same shape.

Table 2. Pre-training.

Property	Value
Pre-training corpus	wood fibre (cellulose)
Training period	more than 2,000 years (from 179–141 BCE)
Architecture fixed	1922 (DIN 476); international since 1975 (ISO 216)
Post-training	fillers and sizing, at the mill
Tokens	not applicable; the model is measured in millimetres
Data contamination	limited to what users write on it

4 Pre-training

AVIVA A4 was pre-trained on wood fibre (Table 2), drawing on more than two thousand years of papermaking, from the earliest surviving paper at Fangmatan to the standard described in §3. The fibres required no supervision: as a sheet dries, hydrogen bonds between neighbouring cellulose fibres hold it together unprompted. Fillers and sizing, added at the mill, are the only post-training.

5 Evaluation

5.1 Hallucination

We define a hallucination as an output containing a false statement. Every mark on a used AVIVA A4 was made by its user; the model adds nothing it was not given. The hallucination rate is therefore 0.00 %. Anything the reader sees in the fibres is their own.

5.2 Refusal

We define refusal as declining to carry a request. AVIVA A4 carried every request written on it, word for word, for as long as the sheet lasted: a refusal rate of 0.0 %. We report this without comment.

5.3 Retrieval

In a needle-in-a-haystack test the needle was an HB pencil, placed on the sheet. Recall was 100 %, falling to 0 % when the pencil was lost.

Table 3. Distillation results.

Model	Pieces	Shape	Capabilities retained	Mass each (calculated)
AVIVA A4 (teacher)	1	$1:\sqrt{2}$	all	4.99 g
AVIVA A5 (student)	2	$1:\sqrt{2}$	all	≈ 2.5 g
AVIVA A6 (student)	4	$1:\sqrt{2}$	all	≈ 1.25 g

5.4 The Arena

Following the practice of blind pairwise comparison, visitors were shown two responses to the prompt “Say something profound” and asked to choose. Both responses were AVIVA A4. AVIVA A4 won 50.0 % of comparisons and lost the other 50.0 % (Figure 3). We consider both results state of the art.

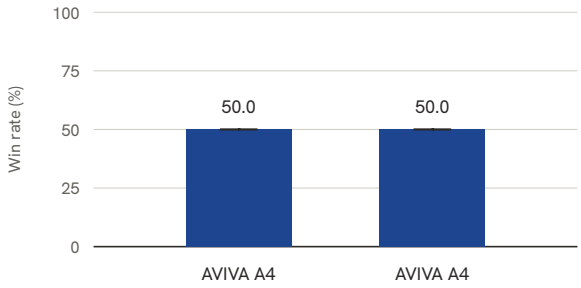


Figure 3. Win rate in blind pairwise comparison. Error bars are present.

5.5 Distillation

We distilled AVIVA A4 by tearing it along its midline. Each student is the same shape as the teacher and retains every capability, including the paper cut (Table 3). A second round produced AVIVA A6. We stopped there, as below A6 the students are better described as confetti.

5.6 Scaling

Each fold doubles thickness [Leading]. Ideally, 42 folds of a 0.1 mm sheet would reach about 440,000 km, beyond the Moon’s average distance of 384,400 km. By Gallivan’s formula [Gallivan], the minimum length needed for n single-direction folds is

$$L = \frac{\pi t}{6}(2^n + 4)(2^n - 1)\,. \tag{2}$$

For $t = 0.1$ mm, seven folds need about 0.88 m, but the long side of A4 is 0.297 m, so six single-direction folds is the ceiling (Figure 4). Her second formula, for folds in alternating directions, permits a seventh; we leave this to future work. The record, 12 folds, was set by Gallivan herself in 2002 using about 1.2 km of tissue paper [Guinness World Records].

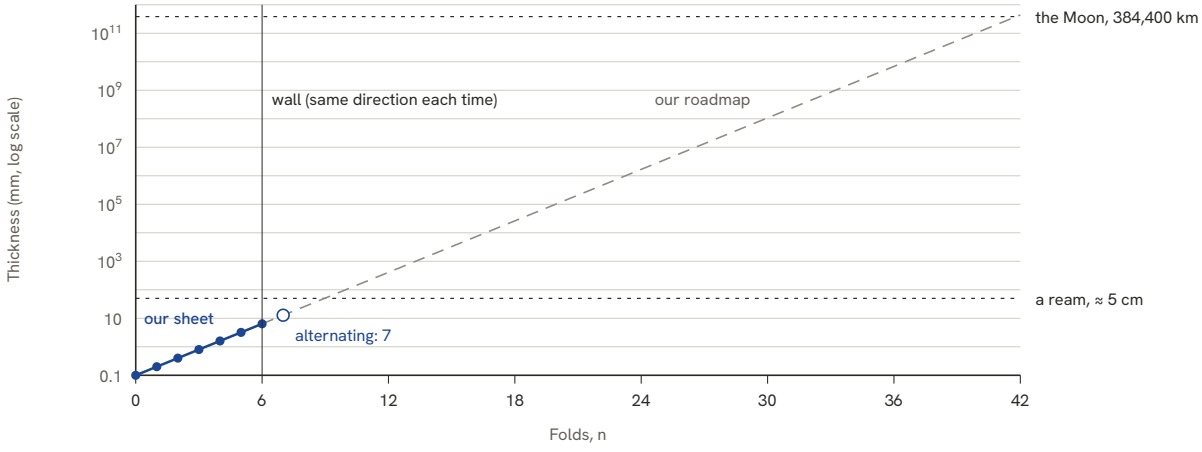
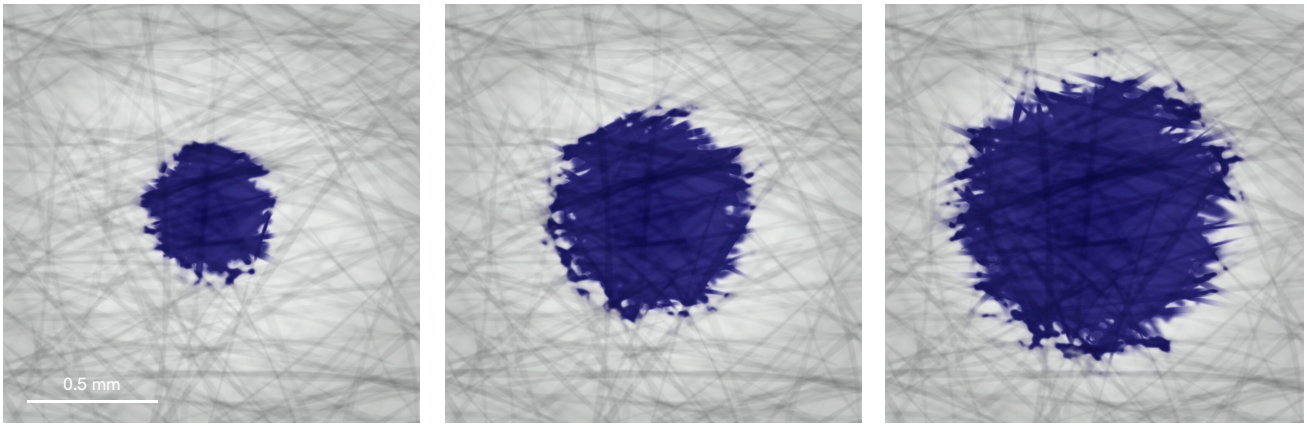


Figure 4. Scaling behaviour for folds made the same way each time. The dashed line is our roadmap. The solid line is our sheet.



(a) t = 0.05 (b) t = 0.35 (c) t = 1.0

Figure 5. Fine-tuning with ink is irreversible. We consider this a feature.

6 Fine-tuning from human feedback

AVIVA A4 is fine-tuned by its user, with a pencil. Graphite rests on the surface of the fibres and is largely removable; ink soaks into the fibres by capillary action and is not [Popular Science; Bleed et al.]. We therefore recommend graphite for experiments and ink for commitments (Figure 5). Erasure is incomplete: the pressure of writing usually leaves an impression after the graphite is gone [Hawkeye Forensic]. In 1770 Joseph Priestley recommended a new gum, sold by Edward Nairne, as "excellently adapted to the purpose of wiping from paper the mark of black-lead-pencil" [Priestley]; it soon came to be called rubber. We have not improved on it.

7 Safety

We are aware of one side effect. The edge of the sheet is microscopically jagged (Figure 6), so it saws rather than slices; fingertips are dense in pain receptors, and the cut is usually too shallow to clot [Big Think; Vallabh]. Paper ignition temperatures quoted in the literature range from about 230 °C to about 450 °C depending on the paper and the method; the widely known figure of 451 °F was popularised by a novel [Bradbury]. We did not test this, and neither should you.

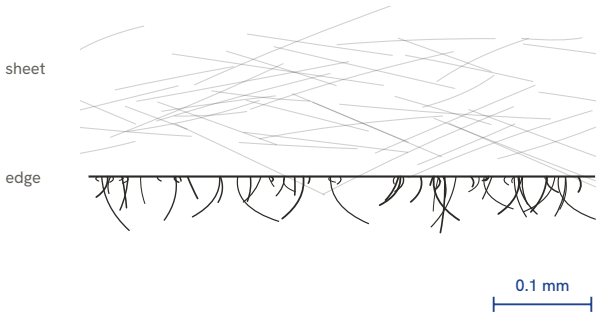


Figure 6. The only known side effect, at magnification.

8 Environmental impact

One life-cycle study of an A4 sheet of office paper estimated 4.29–4.74 g CO_2e per sheet depending on the method, 4.64 g under ISO 14040/44 [Dias & Arroja]. The sheet weighs about 5 g, so the model's footprint is approximately its own weight. Paper fibres are usually said to survive five to seven rounds of recycling before they become too short and stiff to bond, although researchers are aiming for 25 [Karlstad University]; and about three-quarters of the paper and board used in Europe is recycled (75.1 % in 2024) [EPRC]. AVIVA therefore has a finite number of afterlives, which we find reassuring.

9 Limitations

AVIVA A4 knows only what has been written on it. It cannot forget a fold. It will be read by anyone who holds it. Its context window is a hard limit of 210 × 297 mm [Kerning]. A known issue, reported in 1975, predicted that it would be replaced by the paperless office; we are monitoring this.

10 Conclusion

We started with a blank sheet of paper. Like everything else, it now has AI.

Acknowledgements

We thank the pencil, without which this work would be complete.

Table 4. Summary of results.

Metric	AVIVA A4
Hallucination rate	0.00 % (source: the user)
Refusal rate	0.0 %
Needle in a haystack	100 % (pencil present)
Context window	62,370 mm²
Arena win rate	50.0 %
Folds before the wall, same direction each time	6
Folds before the wall, alternating directions	7 (ideal)
Pre-training corpus	wood fibre
Memory	persistent, including erasures

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Appendix A. Response to reviewers

Reviewer 2

"The contribution is unclear. The paper is, however, extremely white."

Response

We thank the reviewer. We have made no changes, as any change would alter the model.

Reviewer 2 (second round)

"Please add more content."

Response

Respectfully, no.

Appendix B. Model listing

`aviva-a4.obj`

```
# AVIVA A4: reference geometry
# 210 x 297 mm, ISO 216. Units: millimetres.
# 4 vertices, 2 faces, 0 opinions.
# Thickness (0.1 mm) omitted for clarity. See the paper, section 3.
o aviva_a4
v 0 0 0
v 210 0 0
v 210 297 0
v 0 297 0
vt 0 0
vt 1 0
vt 1 1
vt 0 1
vn 0 0 1
f 1/1/1 2/2/1 3/3/1
f 1/1/1 3/3/1 4/4/1
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